Section 6.1

Point Estimation

Point Estimation

- Goal: use sample data to estimate an unknown parameter θ (for example μ, σ or p).

- A **point estimate** of θ is a single number computed from the **observed sample** $x_1, \ldots, x_n$. It is a sensible value for θ.
- A **point estimator** $\hat{\theta}$ is the statistic used to get it. It is computed from the **random sample** $X_1, \ldots, X_n$, so it is a random variable.

- Example: $\hat{\mu} = \overline{X}$ is an estimator of μ; $\bar{x} = 12.4$ is an estimate.
- Different estimators can give different estimates, so we need criteria to choose one.

Common Point Estimators

Parameter	Estimator	Estimate
mean μ	$\hat{\mu} = \overline{X}$	$\bar{x}$
variance σ^2	$\hat{\sigma}^2 = S^2$	s^2
standard deviation σ	$\hat{\sigma} = S$	s
proportion p	$\hat{p} = X/n$	x/n

- Estimator: from the random sample (capital letters). Estimate: from the observed sample (lowercase).
- Here X is the number of successes in the n trials.
- Recall: $S^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$.

Example 1: The lifetimes (in hours) of 8 randomly selected batteries are

$$12 \quad 15 \quad 9 \quad 14 \quad 10 \quad 13 \quad 11 \quad 12$$

a) Find a point estimate of the mean lifetime μ.

b) Find point estimates of σ^2 and σ.

c) Find a point estimate of the proportion p of all batteries that last more than 12 hours.

Unbiased Estimators

- A point estimator $\hat{\theta}$ is **unbiased** if

$$E(\hat{\theta}) = \theta \quad \text{for every value of } \theta.$$

- The **bias** of $\hat{\theta}$ is $E(\hat{\theta}) - \theta$.

- Unbiased estimators we already know:
  - $\overline{X}$ for μ, since $E(\overline{X}) = \mu$;
  - $\hat{p} = X/n$ for p, since $E(X) = np$;
  - S^2 for σ^2. This is why S^2 divides by $n - 1$.
- S is **not** unbiased for σ, but its bias is small.
- Among unbiased estimators, prefer the one with the smallest variance.

Example 2: Let X_1, X_2, X_3 be a random sample from a distribution with mean μ and variance σ^2. Consider

$$\hat{\theta}_1 = \frac{X_1 + X_2 + X_3}{3}, \quad \hat{\theta}_2 = \frac{X_1 + 2X_2 + X_3}{4}, \quad \hat{\theta}_3 = \frac{X_1 + X_2}{3}.$$

a) Which of these are unbiased estimators of μ? Find the bias of any that are not.

b) Among the unbiased ones, which would you prefer? Why?